

Lab Report PDSD:

LOSS OF STABILITY TEST (ASTM D6927)

Objective

To evaluate the Marshall stability of asphalt specimens before and after conditioning and determine the retained stability. The results are compared against the acceptance criterion that the **Loss of Stability (LOS) should be less than 20%**.

Apparatus

- Marshall Stability testing machine
- Load gauge (least count = 0.001 inch)
- Deformation gauge (least count = 0.001 mm)
- Water bath for conditioning

Procedure (Summary)

1. Prepare compacted Marshall specimens according to ASTM D6927.
2. Measure stability of one specimen without conditioning (S_1).
3. Condition another specimen in water bath as per standard, then measure stability (S_2).
4. Calculate percent stability retained and percent loss.

Results

- Without conditioning (S_1): 1015
- With conditioning (S_2): 680

Calculations:

$Stability\ Retained = S_2 / S_1 \times 100 = 680 / 1015 \times 100 = 67\%$ $Loss\ of\ Stability\ (LOS) = 100 - 67 = 33\%$

Results Table

Sample No.	Deflection / Flow (mm)	Load (N)	Average Stability (N)	IRS (%)	LOS (%)	Remarks
1 (Conditioned)	110	560	680	67	33	Sample is not Satisfied the criteria.
2 (Conditioned)	140	800				
3 (Unconditioned)	160	1219	1015			
4 (Unconditioned)	100	810				

Discussion

- The specimen retained **67%** of its original stability after conditioning.
- The **loss of stability is 33%**, which exceeds the acceptable limit of **20% (LOS < 20%)** for durable mixes.
- This indicates that the tested asphalt mixture is **not satisfactory in moisture resistance** and may fail under field conditions.

Conclusion:

The Marshall stability test showed a **33% loss of stability**, higher than the allowable **20%**. Therefore, the tested mix **does not meet the standard durability requirement**.